Chapter 6.3 — Formula Sheet

Combinatorial & Two-Player Game Theory

Companion to your Notebook 6 syllabus, your *21 Worked Examples* (Ex 1–21) and the *Stage-1 Hard Drill* (Q1–Q10)

A. P/N Positions

Normal Play Labelling [Ex 1–2; Q1]

Player who cannot move loses. Terminal positions are P.

$$P: \text{ all moves} \to N \qquad N: \text{ some move} \to P$$

P = previous player wins (mover loses). Fill the table upward from $n = 0$.

Subtraction Games: Periods [Ex 1–4; Q1]

Moves $\{1, \ldots, k\}$: $P \iff (k{+}1) \mid n$ (reply $k{+}1-m$). $\{1,3,4\}$: $n \equiv 0, 2 \pmod 7$. $\{2,5,6\}$: $n \equiv 0, 1, 4, 8 \pmod{11}$. Any power of 2: $P \iff 3 \mid n$. Largest move $k \Rightarrow$ once k consecutive entries repeat, the pattern is periodic. **Misère** $\{1,2,3\}$: mover loses iff $n \equiv 1 \pmod 4$, one off from normal play.

Invariants and Parity [Ex 5–6]

Chocolate bar $m \times n$ (break one piece per move): exactly $mn - 1$ moves, so first player wins iff mn is even. Grid race (left/down/diagonal by 1 to $(0,0)$): $P \iff$ both coordinates even. Look for a fixed game length before building a table.

B. Nim and XOR

Bouton's Theorem [Ex 7–10; Q3]

Nim-sum $s = a_1 \oplus \cdots \oplus a_k$ (bitwise XOR):

$$P \iff s = 0$$

If $s \neq 0$: pick a heap with the *leading bit of* s set and replace $a \to a \oplus s < a$. Heaps without that bit cannot move to 0. Examples: $(1,2,3)$ is P; $(3,5,7)$: s=1, three winning moves; $(5,8,11,14)$: s=8, three; $(13,21,34,55)$: s=13, remove the heap of 13.

Why It Works [Ex 8, 10]

From $s = 0$ any move changes one heap $a \to a'$, new sum $= a \oplus a' \neq 0$. From $s \neq 0$ the leading column has odd parity, so an odd number of heaps contain it: with 3 heaps there are 1 or 3 winning moves, never 2 [Q10].

Counting with XOR [Ex 11; Q10]

Heaps iid on $\{0, \ldots, 2^k - 1\}$, ≥ 2 heaps: $\mathbb{P}(s = 0) = 2^{-k}$. Three heaps on $\{1, \ldots, 7\}$: $\#P = 7 \cdot 6 = 42$, $\mathbb{P} = \frac{6}{49}$; four heaps: $\frac{301}{2401} = \frac{43}{343}$. $\mathbb{E}[\#\text{win moves}] = \frac{525}{343} = \frac{75}{49}$.

C. Variants and Grundy Values

Sprague–Grundy [Ex 13; Q2, Q6]

$$G(\text{position}) = \operatorname{mex}\{G(\text{options})\}, \qquad G(A + B) = G(A) \oplus G(B)$$

Single game is $P \iff G = 0$. Subtraction $\{1,2,3\}$ (and Nim capped at 3 per move): $G(n) = n \bmod 4$, so use $a_i \bmod 4$ in the XOR. Grundy's game (split into unequal parts), $n = 1..9$:

$$0, 0, 1, 0, 2, 1, 0, 2, 1$$

A winning move changes one component's G to $G \oplus s$.

Wythoff and Misère Nim [Ex 12, 14; Q4, Q5]

Wythoff (one heap, or equal from both):

$$P\text{-pairs } (a_k, a_k{+}k) = (\lfloor k\varphi \rfloor, \lfloor k\varphi^2 \rfloor)$$

$(0,0), (1,2), (3,5), (4,7), (6,10), (8,13), (9,15), (11,18), (12,20), \ldots$ XOR rule fails. 23 ordered P-positions with $a, b \leq 30$. **Misère Nim:** play normal Nim while ≥ 2 heaps stay > 1; if one heap > 1 remains, leave an *odd* number of 1-heaps. $(1,1,1,4) \to 4 \to 0$; $(1,1,1,1,5) \to 5 \to 1$.

D. Matrix Games

Maximin, Minimax, Saddle Points [Ex 15–16; Q7]

Row maximises A_{ij}:

$$\underline{v} = \max_i \min_j A_{ij} \;\leq\; \overline{v} = \min_j \max_i A_{ij}$$

Equal $\Rightarrow$ saddle point (min of its row, max of its column), value $= \underline{v}$. **Minimax theorem:** with mixed strategies $\underline{v} = \overline{v} = v$. Value is unique, saddle points need not be. **Dominance:** delete a row that is $\leq$ another row everywhere, a column that is $\geq$ another column everywhere; repeat.

2×2: Indifference Principle [Ex 17–18; Q7]

$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, no saddle, $D = a + d - b - c$:

$$p = \frac{d-c}{D}, \quad q = \frac{d-b}{D}, \quad v = \frac{ad-bc}{D}$$

p = prob. of row 1, q = prob. of column 1. Equalise the *opponent's* payoffs. $\begin{pmatrix} 3 & -1 \\ -2 & 4 \end{pmatrix}$: $p = \frac{3}{5}, q = \frac{1}{2}, v = 1$. Fingers (even-sum wins): $p = q = \frac{7}{12}$, $v = -\frac{1}{12}$. Parametric $\begin{pmatrix} x & 2 \\ 3 & 1 \end{pmatrix}$: $x \geq 2$ saddle v=2; $x < 2$: $v = \frac{6-x}{4-x}$.

$2 \times n$ Graphical Method [Ex 19]

Row plays row 1 w.p. p: each column gives a line in p. Row maximises the *lower envelope*; the peak is where a rising and a falling line meet. Column then mixes just those two columns. $\begin{pmatrix} 1 & 3 & -2 \\ -1 & 0 & 4 \end{pmatrix}$: $p = \frac{5}{8}$, $v = \frac{1}{4}$, $q = (\frac{3}{4}, 0, \frac{1}{4})$.

Special Structures [Ex 21; Q8]

Diagonal $\operatorname{diag}(v_1, \ldots, v_n)$, $v_i > 0$ (hide and seek):

$$V = \frac{1}{\sum_i 1/v_i}, \qquad x_i = y_i = \frac{V}{v_i}$$

Values $1,2,3$: $V = \frac{6}{11}$; $1,2,4$: $V = \frac{4}{7}$. 2×2 diagonal: $v = \frac{\alpha\beta}{\alpha+\beta}$. **Skew-symmetric** ($A^T = -A$, symmetric game): $v = 0$, identical optimal strategies, solve $Aq = 0$, $\sum q = 1$. Weighted RPS $\left(\begin{smallmatrix} 0 & -2 & 1 \\ 2 & 0 & -3 \\ -1 & 3 & 0 \end{smallmatrix}\right)$: $q = (\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$.

Bluffing Game [Ex 20; Q9]

Ante 1, bet b, High/Low $\frac{1}{2}$ each: Honest/Bluff against Call/Fold gives $\begin{pmatrix} b/2 & 0 \\ 0 & 1 \end{pmatrix}$:

$$v = \frac{b}{b+2}, \quad p(\text{Honest}) = q(\text{Call}) = \frac{2}{b+2}$$

b=1: $v = \frac{1}{3}$, $p = q = \frac{2}{3}$. b=3: $v = \frac{3}{5}$. Against an always-call opponent Honest earns $\frac{b}{2}$.

E. 60-Second Recipe

Any Game Question

Sequential, last mover wins: look for an invariant; else table P/N and find the period; for heaps use XOR; for unfamiliar moves compute Grundy values. **Simultaneous:** compute $\underline{v}, \overline{v}$; delete dominated lines; 2×2 formulas; check by testing the opponent's pure replies.

Reminder: P means the player about to move loses; $s = 0$ is the whole of Nim; Grundy values turn every impartial game into Nim; mixed strategies equalise the opponent's payoffs, never your own; always check the answer against each pure reply.